

Workflow reliability — Build plan

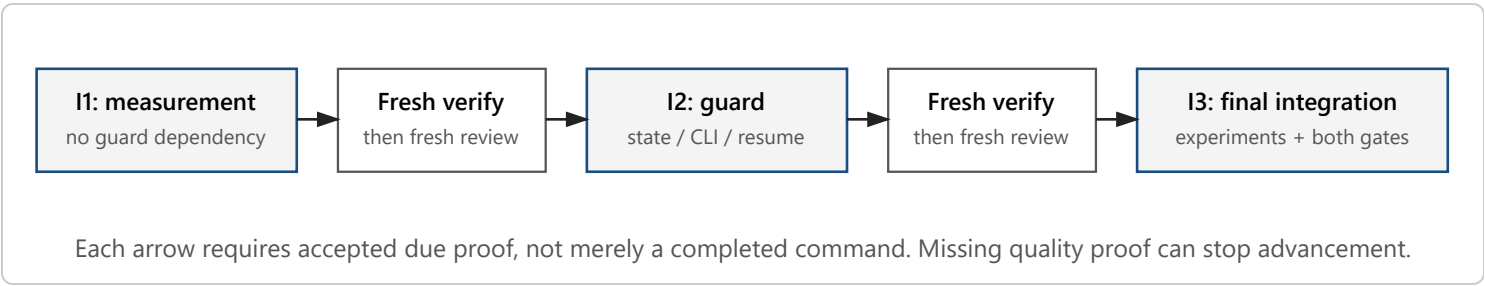
Planning r1 · 16 September 2026 · 11 tasks / 3 increments
Authority: [AC01-09 and exclusions](#) · [research](#) · [approved design r2](#) · [r2 contracts](#) · [independent approval](#)
Execution detail: [tasks.json](#) owns task dependencies, commands, assertions and gate schedules. [Traceability](#) remains an evidence index, not a second plan.

1. Outcome and current boundary

Build standalone, honest ESM/CJS measurements first; then real guarded admission and durable resume; then canonical workflow guidance and separate fresh-agent experiments. **FACT:** planning HEAD is `dde2b491` ; differences from research base `bcc971d3` are task artifacts/assets only. Approved r2 hashes match the independent review. Existing native baseline is **51/51**, not new-feature proof.

Parent accepted r2 under authorized autopilot; human approval is unconfirmed. Every task and feature check below is planned, not executed. No implementation, installation, agent launch, commit or publication belongs to this planning handoff.

2. Build order



Increment / tasks	Owned design changes and files	ACs / prerequisites / proof
□ I1 · P0 I1.T1–T4	<code>evidence.py</code> ; native reporter, mutation and runner <i>env only</i> ; probe and scorer; their tests and metric-consumer sections.	01, 02, 09; partial 06. Approved r2 → primitives → native/probe → integrated real Git CLI fixtures, native/eval regression, fresh verifier and reviewer.
□ I2 · P0/P1 I2.T1–T4	<code>workflow_state.py</code> , <code>workflow_gate.py</code> , <code>workflow.py</code> ; runner observer; CLI reference; Python/native workflow failure tests. Runtime current-design, research binding and temporal rules belong here.	03–09 runtime scenarios. I1 gate → state/readiness → dispatch → negative/positive public CLI pairs, live-child recovery and fresh-process resume → both independent gates.
□ I3 · P1/P2 I3.T1–T3	Remaining SKILL/reference consumers; conditional live-bundle staging; parent-host resume and held-out diagnosis evidence; final report/index reconciliation.	04–09 integration; final 01–09. I2 gate → canonical wiring → neutral-input experiments → all regressions, fresh verifier, fresh reviewer and truthful final status.

3. Safe implementation and proof

First ready task: I1.T1. Confirm the r2 source-snapshot/atomic-write/lock contracts, add independent primitives and their real filesystem tests. After its tests and shared-contract handoff, I1.T2 and I1.T3 may run concurrently: one implementer owns mutation/native/scorer/runner-env; parent owns probe/metric docs. No shared-file concurrent edits. Parent integrates both before I1.T4. Otherwise execute sequentially.

Every increment runs unit, integration and public CLI E2E checks, then a **fresh non-implementing verifier and a separate fresh read-only reviewer before any commit or next increment**. Rework invalidates affected proof and is reverified/reviewed. Exact argv, Windows cwd, bounds, positive assertions, cleanup, outcome owners and evidence locations are in the linked command/check records. Native 22+ is the intended compatibility contract; only Node 24.11.1 is available here. Do not claim a Node 22 test.

4. Honest completion and recovery

Functional tests proving *incomplete/fail* are expected to pass; an actual default probe lacking required collectors is expected to fail. These are different verdicts. No fixture supplies missing real measurements. Required unavailable coverage/architecture proof blocks complete quality/guard closure; the plan grants no waiver or automatic onward transition. Preserve reviewed functional work and report the blocker rather than fabricate a green increment.

Due checks are listed only at each gate task. Later checks stay unverified; spanning ACs stay partial. Changes reopen bound dependents, retaining unaffected receipt IDs and historical pre-action proof. Unknown spawn/tree state stays locked: guard death, age and PID absence cannot establish recovery. Record timeouts; verify owned cleanup; retry once after a concrete remedy, then stop affected work.

5. Final evidence and exclusions

Final regression repeats all Python, native, fixed-corpus and dry-run checks plus all due scenarios. Parent-host resume and diagnosis use neutral input/history manifests and a separate evaluator; capture accesses and disclose shared storage, never claim a sandbox. Exposure or insufficient records leaves blindness unverified. Full quality needs real measurements, not optimistic summaries.

Preserve historical planner evidence, generic mutation 60%, eval 0.9 and null normalization. No new framework, measurement exemptions, authenticated identities or temporary-location-only refactor. Production deployment is N/A: local CLI/docs delivery. GitHub publication remains 403/EMU blocked; do not retry or bypass. Parent owns Gate 3 consumer readiness and later state/traceability updates. Printed/render verification of this new plan remains a parent handoff check; no page-count measurement is claimed.